

United States Top 50 Playlist Performance and Song Popularity Trend Analysis

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Abstract

This project analyzes Spotify United States Top 50 playlist data to identify trends in song popularity, artist dominance, playlist longevity, and ranking behavior. An interactive Streamlit dashboard was developed using Python, Pandas, and Plotly.

Introduction

Music streaming platforms generate large amounts of data every day. Understanding playlist trends helps identify popular songs, artists, and listening patterns.

Objectives

Analyze song popularity trends, identify dominant artists, measure playlist longevity, compare explicit and non-explicit songs, and visualize ranking patterns.

Dataset Description

The dataset contains date, position, song title, artist name, popularity score, duration, album type, track count, and explicit-content information. The dataset consists of approximately 27,800 records.

Methodology

Data preprocessing was performed using Pandas. Exploratory Data Analysis was conducted and visualizations were built using Plotly and Streamlit.

Results and Findings

The analysis identified highly popular songs, dominant artists, and patterns in playlist persistence. Popularity showed a strong relationship with playlist position.

Technologies Used

Python, Pandas, Plotly, Streamlit, GitHub

Conclusion

The dashboard successfully demonstrates how data analytics can be applied to music streaming data and provides useful insights through interactive visualizations.